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Dynamic Material Balance (Oil or Gas-In-Place Without Shut-Ins)

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Abstract

Material Balance calculations for determining oil- or gas-in-place are based on obtaining static reservoir pressures as a function of cumulative production. This requires the wells to be shut-in, in order to determine the average reservoir pressure. In a previous publication[1], it was shown that the material balance calculation could be done without shutting-in the well. The method was called "Flowing Material Balance". While this method has proven to be very good, it is limited to a constant flow rate, and fails when the flow rate varies.

The "Dynamic Material Balance" is an extension of the Flowing Material Balance. It is applicable to either constant flow rate or variable flow rate, and can be used for both gas and oil. The "Dynamic Material Balance" is a procedure that converts the flowing pressure at any point in time to the average reservoir pressure that exists in the reservoir at that time. Once that is done, the classical material balance calculations become applicable, and a conventional material balance plot can be generated.

The procedure is graphical and very straightforward: a) knowing the flow rate and flowing sandface pressure at any given point in time, convert the measured flowing pressure to the average pressure that exists in the reservoir at that time; b) use this calculated average reservoir pressure and the corresponding cumulative production, to calculate the original oil- or gas-in-place by traditional methods. The method is illustrated using data sets.

Introduction

The material balance method is a fundamental calculation in reservoir engineering, and is considered to yield one of the more reliable estimates of hydrocarbons-in place. In principle, it consists of producing a certain amount of fluids, measuring the average reservoir pressure before and after the production, and with knowledge of the PVT properties of the system, calculating a mass balance as follows:

Remaining Hydrocarbons-in-place = Initial Hydrocarbons-in-place – Produced Hydrocarbons

At face value, the above equation is simple; however in practice, its implementation can be quite complex, as one must account for such variables as external fluid influx (water drive), compressibility of all the fluids and of the rock, hydrocarbon phase changes, etc…

In order to determine the average reservoir pressure, the well is shut-in, resulting in loss of production. In high permeability reservoirs, this may not be a significant issue, but in medium to low permeability reservoirs, the shut-in duration may have to last several weeks (and sometimes months) before a reliable reservoir pressure can be estimated. This loss of

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production opportunity as well as the cost of monitoring the shut-in pressure is often unacceptable.

It is clear that the production rate of a well is a function of many factors such as permeability, viscosity, thickness etc… Also, the rate is directly related to the driving force in the reservoir, i.e. the difference between the average reservoir pressure and the sandface flowing pressure. Therefore, it is reasonable to expect that knowledge about the reservoir pressure can be extracted from the sandface flowing pressure if both the flow rate and flowing pressure are measured. If, indeed, the average reservoir pressure can be obtained from flowing conditions, then material balance calculations can be performed without having to shut-in the well. This is of great practical value.

In a previous publication[1] the authors presented "The Flowing Material Balance" for gas wells flowing at a constant rate. Experience has shown that this method works very well, but unfortunately is limited to cases where the well is flowing at a constant rate. The following development extends the Flowing Material Balance method to cases where the flow rate is not constant. It is called the Variable Rate Flowing Material Balance or "Dynamic Material Balance". This name has been chosen to contrast with the traditional material balance procedure, which relies on "static" reservoir pressure data.

A review of the Flowing Material Balance method (constant flow rate) is given below to introduce the concepts of the method. This is then followed by development of the Dynamic Material Balance by extending the constant rate analysis to the variable rate situation, thus generalizing the applicability of the method.

For the purposes of this paper, the equations are derived for a "volumetric" reservoir (i.e. no water drive or external fluid influx), but the method can be extended to include such complexities. The method is valid for both oil and gas systems, but it is sometimes more convenient to present a particular concept (or equation) in terms of gas rather than oil, or vice versa.

Flowing Material Balance

Strictly speaking, both the Flowing Material Balance (constant rate) and the Dynamic Material Balance (variable rate) are valid only when the flow has reached "Boundary Dominated" conditions. The principles underlying these methods are best illustrated using **constant rate** production. When the flow becomes dominated by the boundaries, i.e. stabilized or "pseudo-steady-state" conditions are achieved, the pressure at every point in the reservoir declines at the same rate. This is illustrated in Figure 1, which shows that the pressure drop measured at the wellbore is the same as the pressure drop that would be observed anywhere in the reservoir, including the location which represents average reservoir pressure. p_{R1}, p_{R2} and p_{R3} represent the average (static) reservoir pressure that would be obtained if the well was shut-in at times *t1*, *t2*, and *t3*. It is evident, from Figure 1, that the change in average reservoir pressure is equal to the change in the sandface flowing pressure.

$$\bar{p}_{R1} - \bar{p}_{R2} = pwf_1 - pwf_2 \quad (1)$$

$$\bar{p}_{R2} - \bar{p}_{R3} = pwf_2 - pwf_3 \quad (2)$$

Rearranging,

$$\bar{p}_{R1} - pwf_1 = \bar{p}_{R2} - pwf_2 = \bar{p}_{R3} - pwf_3 \quad (3)$$

Thus, if the sandface flowing pressure and the average reservoir pressure are plotted versus time (or cumulative production), they will have the same trend, and will be displaced by a constant. In a conventional material balance calculation, reservoir pressure is measured or extrapolated based on stabilized shut-in pressures at the well. While a well is flowing, it is obvious that the average reservoir pressure cannot be measured, but the equations above give the relationship between the well flowing pressure (which can be measured) and the average reservoir pressure.

Constant Rate Flowing P/Z Plot

Appendices A, B and C develop the equations that relate average reservoir pressure to flowing pressure. For a gas reservoir, the equations are given in terms of pseudopressure, and the material balance is expressed in terms of *p/z*.

Figure 2 demonstrates the Flowing Material Balance as applied to a gas reservoir. It shows how the flowing pressure (p_{wf}/z) and the average reservoir pressure (p_R/z) are related, and how the Original-Gas-In-Place (OGIP) can be obtained from the flowing pressure if the initial pressure is known. The line drawn through the measured flowing pressure data needs only to be "shifted" upwards so that it goes through the initial (p_i/z_i) point.

Dynamic Material Balance (Variable Rate Flowing P/Z Plot)

The Flowing Material Balance described above has proven to be a very successful way of determining original-gas-in-place when the flow rate is held constant. However it fails completely if the flow rate is variable. Unfortunately most wells do not flow at constant rate for extended periods of production. A typical high deliverability gas well may have a production profile as shown in Figure 3.

A different methodology, called the Dynamic Material Balance, has been developed, and is the subject of this paper. It is applicable to both constant rate and variable rate production. It is obvious that, for the flowing pressure profile seen in Figure 3, we cannot assume a constant pressure difference between the average reservoir pressure and the measured flowing pressure. The complete development of the appropriate equations is given in Appendices A, B and C, but a simplified summary of the concepts as they apply to variable rate production is summarized below:

Pseudosteady State Flow:

$$p_i - p_{wf} = \frac{qt}{c_o N} + b_{pss} q \quad (4)$$

Cumulative Production:

$$(q \times t = N_p) \quad (5)$$

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Material Balance Equation:

$$p_i - \overline{p}_R = \frac{N_p}{c_o N} \quad (6)$$

Combing equations 4, 5 and 6:

$$\overline{p}_R - p_{wf} = b_{pss} q \quad (7)$$

Re-arranging:

$$\overline{p}_R = p_{wf} + b_{pss} q \quad (8)$$

The above equation illustrates how the Dynamic Material Balance can be applied to a well with varying production rate and correspondingly varying flowing pressure. The conversion from flowing pressure to average reservoir pressure must take into account the varying flow rate. Since the flow rate is known, we need only determine the value of b_{pss} , using some independent method. One way to obtain a reliable estimate of b_{pss} is discussed in Appendix A. A plot of (p_i-p_{wf}/q) versus N_p/q should yield a straight line when boundary dominated flow is reached. The intercept of this plot is b_{pss} . Note that the value of b_{pss} is subject to interpretation, as it depends on the proper identification of the stabilized (straight-line) section of the graph.

The above summary equations are for a single phase liquid system. The corresponding equations for a gas reservoir are developed in Appendix C.

For a gas reservoir, two modifications are necessary:

a) The pressure must be converted to pseudopressure, p_p, to account for the dependence of viscosity (μ) and Z-factor on pressure, and

b) material-balance-time must be converted to pseudotime, t_{ca}, to account for the strong dependence of gas compressibility, c_g, on pressure.

The step by step procedure for generating a Dynamic Material Balance plot for a gas well with varying flow rate is given below:

1. Convert initial pressure to pseudopressure, p_{pi}
2. Convert all flowing pressures to pseudopressures, p_{pwf}
3. Assume a value for the Original Gas in Place, G
4. Calculate pseudotime from Equation C-11
5. Plot $(p_{pi}-p_{pwf}/q)$ versus pseudotime, $t_{ca \cdot s}$. The intercept gives b_{pss}. See Figure 4.
6. Calculate the average reservoir pseudopressure from Equation C-19.
7. Convert the average reservoir pseudopressure to average reservoir pressure, p_R.
8. Calculate p_R/Z and plot against cumulative gas produced, Gp, just like the conventional Material Balance graph for a gas pool. The intercept on the X-axis gives the original-gas-in-place, G. See Figure 5.
9. Using this new value of G, repeat steps 3 to 7 until G converges. See Figure 5

Limitations

The procedures described in this paper are very effective and provide extremely valuable information. However, like any other reservoir engineering, it has its limitations.

- Because the formulation of the material balance time and pseudotime are, strictly speaking, rigorous only during boundary-dominated flow, data obtained during transient flow cannot be used in this analysis. However, for the majority of production data, this is not a problem. The transient data can be identified as the curved part of the graph in Figure 4 and should be ignored.
- Experience with this method has shown that in certain situations such as pressure-dependent permeability, or continuously changing skin, (both factors have been ignored in the development of the equations) this method will tend to under-predict the hydrocarbons-in-place. However, these factors can readily be accounted for by more complex definitions of pseudopressure and pseudotime.
- When comparing the Dynamic Material Balance to the more traditional build-up tests for obtaining average reservoir pressure, it should be kept in mind that both methods have their strengths and their limitations. The dynamic material balance is an "indirect" method of determining the average reservoir pressure. As such, it incorporates many assumptions. On the other hand, buildup tests themselves have their own sets of assumptions when the buildup pressure has to be extrapolated to obtain the average reservoir pressure. Accordingly, whenever possible, these methods should be used in concert with each other rather than as alternatives to each other.

Conclusion

- It is possible to obtain the average reservoir pressure without shutting-in a well.
- The flowing pressure can be converted to the average reservoir pressure existing at the time of the measurement using a very simple and direct procedure.
- The average reservoir pressure obtained from the Dynamic Material Balance method can be used anywhere it is traditionally used.
- For a gas well, a conventional p_R/Z plot can easily be generated without shutting-in the well, and the original-gas-in-place determined as usual.
- The Dynamic Material Balance applies to variable rate production. It is an extension of the Flowing Material Balance method which was limited to a constant rate situation.
- The Dynamic Material Balance should not be viewed as a replacement to buildup tests, but as a very inexpensive supplement to them.

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NOMENCLATURE

A = Reservoir area, ft^2

B = Formation volume factor, bbl/stb

b_{pss} = Reservoir constant (Equation A-4)

$\bar{c}_g$ = Gas compressibility at average reservoir pressure, psi^{-1}

c_{gi} = Gas compressibility at initial reservoir pressure, psi^{-1}

c_o = Oil compressibility, psi^{-1}

G = Original gas in place, MMscf

G_p = Cumulative gas produced, MMscf

h = Pay thickness, ft

k = Reservoir permeability, md

N = Original oil in place, Bbl

N_p = Cumulative production produced, Bbl

p_D = Dimensionless pressure, $\frac{(p_i - p)kh}{141.2qB\mu}$ or $\frac{(p_{pi} - p_p)kh}{1.417\times10^6 qT}$

p_i = Initial reservoir pressure, psi

$\bar{p}_R$ = Average reservoir pressure, psi

p_{st} = Standard pressure, (14.65 psi in Alberta)

p_{wf} = Flowing pressure at the interface, psi

p_p = Pseudopressure, (Equation C-2)

$\bar{p}_p$ = Pseudo-pressure corresponding to average reservoir pressure $\bar{p}$, psi^2/cp

$\bar{p}_{pD}$ = Dimensionless pseudo-pressure difference corresponding to average reservoir pressure, $\frac{(p_{pi} - p_p)kh}{1.417\times24qT}$

p_{p_i} = Pseudo-pressure corresponding to initial reservoir pressure, psi^2/cp

p_{pwf} = Pseudo-pressure corresponding to the sandface flowing pressure, psi^2/cp

q = Production rate (can be a function of time),BPD or MMscfd

r_e = Exterior radius, feet

r_{eD} = Exterior radius dimensionless, $\frac{r_e}{r_w}$

r_{wa} = Apparent wellbore radius, feet

r_w = Wellbore radius, feet

t = Time, day

t_a = Pseudo-time, day-psi/cp

t_c = Material balance time for liquid, day

t_{ca} = Material balance pseudo-time for gas (Equation C-11), day

t_D = Dimensionless time, $\frac{2.637\times10^{-4}kt\times24}{\phi\mu crw^2}$

T = Reservoir temperature, R°

T_{st} = Standard temperature, 519.668 R°

$\bar{Z}$ = Gas compressibility factor at average reservoir pressure

Z_i = Gas compressibility factor at initial reservoir pressure

ϕ = Hydrocarbon filled porosity

μ = Viscosity, cp

μ_i = Viscosity at initial reservoir pressure, cp

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Appendices

Appendix A:

Flowing Material Balance: (Constant Rate) Oil:

The pseudosteady state equation for an oil well, above the bubble point, flowing at a constant rate is given by Lee[3]:

$$p_D = 2t_D / r_{eD}^{\,2} + \ln(r_{eD}) - \frac{3}{4} \quad \text{(A-1)}$$

This translates to:

$$p_i - p_{wf} = \frac{qt}{c_o N} + \frac{141.2 q B\mu}{kh}\left[(\ln\frac{r_e}{r_{wa}}) - \frac{3}{4}\right] \quad \text{(A-2)}$$

$$p_i - p_{wf} = \frac{qt}{c_o N} + b_{pss} \times q \quad \text{(A-3)}$$

where,

$$b_{pss} = \frac{141.2 B\mu}{kh}\left[\ln(\frac{r_e}{r_{wa}}) - \frac{3}{4}\right] \quad \text{(A-4)}$$

Note that bpss is a constant. The form of this equation was given in Blasingame(2).

Recognizing that in Equation A-3, the term qt is the cumulative production, Np. The cumulative production relates the initial reservoir pressure to the current reservoir pressure through the Material Balance Equation for an oil reservoir above the bubble point:

$$p_i - \overline{p}_R = \frac{N_p}{c_o N} = \frac{qt}{c_o N} \quad \text{(A-5)}$$

Combining Equations A-3 and A-5

$$\overline{p}_R - p_{wf} = b_{pss} \times q \quad \text{(A-6)}$$

$$\overline{p}_R = p_{wf} + b_{pss} \times q \quad \text{(A-7)}$$

This equation shows that if b_{pss} were known, the average reservoir pressure at any time can be determined by measuring the flowing pressure and simply adding to it the term b_{pss} x q , where q is the instantaneous flow rate.

bpss can be determined by rearranging Equation A-3 as follows:

$$\frac{(p_i - p_{wf})}{q} = \frac{qt}{c_o N q} + b_{pss} = \frac{N_p}{c_o N q} + b_{pss} \quad \text{(A-8)}$$

A Cartesian plot of $(p_i - p_{wf}/q)$ versus N_p/q will yield a straight line with an intercept of b_{pss}.

Appendix B:

Dynamic Material Balance: (Variable rate) Oil:

Strictly speaking, the relationships developed in Appendix A apply to a constant rate situation only.

Numerous publications[5][6][7] in the field of production data analysis have demonstrated that if the flow time, t, is replaced by Material-Balance-Time, t_c, the equations of Appendix A are valid for varying rate production. For an oil reservoir, t_c is defined as:

$$t_c = \frac{N_p}{q} \quad \text{(B-1)}$$

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Accordingly, for any flow condition (constant rate or variable rate) the analysis procedure is:

a) Plot a Cartesian graph of $(p_i - p_{wf}/q)$ versus N_p/q. The early part of the data may be curved because of transient flow. However, the boundary-dominated flow will yield a straight line with an intercept equal to b_{pss}.

b) Convert the measured flowing pressure to the average reservoir pressure existing in the reservoir at that time using Equation A-7

$$p_R = p_{wf} + b_{pss} \times q \quad \text{(A-7)}$$

Appendix C:

Dynamic Material Balance: (Variable Rate)

Gas:

The development of the equations for gas flow parallels that for oil flow (Appendix A).

$$p_D = \frac{2t_D}{r_{eD}^2} + \ln(r_{eD}) - \frac{3}{4} \quad \text{(A-1)}$$

Substituting for the dimensionless quantities in terms of gas variables (ERCB 1975, equation 4N21):

$$p_{pi} - p_{pwf} = \frac{24 \times 2348 \times T \times q \times t}{\pi \times \phi \times \mu_i \times c_{g_i} \times r_e^2 \times h} + \frac{1.417 \times 10^6 \times q \times T}{k \times h} \times \left[\ln\left(\frac{r_e}{r_{wa}}\right) - \frac{3}{4} \right] \quad \text{(C-1)}$$

where pseudopressure, p_p is defined by:

$$p_p = 2\int \frac{p}{\mu Z} dp \quad \text{(C-2)}$$

In the same manner as for the oil equations in Appendix A, the Material Balance Equation for gas will be incorporated into Equation C-1.

The gas material balance can be stated as

$$\frac{\bar{p}}{\bar{Z}} = \frac{p_i}{Z_i}\left(1 - \frac{G_p}{G}\right) \quad \text{(C-3)}$$

Differentiating partially with respect to real time, t, one gets

$$\frac{\partial}{\partial t}\left(\frac{\bar{p}}{\bar{Z}}\right) = -\frac{p_i q}{Z_i G} \quad \text{(C-4)}$$

where $q(t) = \dfrac{dG_p(t)}{dt}$ (C-5)

Similarly from partially differentiating Equation (C-2) with respect to $\bar{p}$, one gets

$$\frac{\partial \bar{p}_p}{\partial \bar{p}} = \frac{2\bar{p}}{\mu \bar{Z}} \quad \text{(C-6)}$$

One can also recognize that

$$\frac{\partial}{\partial \bar{p}}\left(\frac{\bar{p}}{\bar{Z}}\right) = \frac{1}{\bar{Z}} - \frac{p}{\bar{Z}^2}\frac{d\bar{Z}}{dp} = \frac{\bar{p}\,\bar{c}_g}{\bar{Z}} \quad \text{(C-7)}$$

where the gas compressibility is defined as

$$\bar{c}_g = \frac{1}{\bar{p}} - \frac{1}{\bar{Z}}\frac{\partial \bar{Z}}{\partial \bar{p}} \quad \text{(C-8)}$$

Now, using the chain rule

$$\frac{\partial \bar{p}_p}{\partial t} = \frac{\partial}{\partial t}\left(\frac{\bar{p}}{\bar{Z}}\right) \cdot \frac{\partial \bar{p}_p}{\partial \bar{p}} \cdot \left[\frac{\partial}{\partial \bar{p}}\left(\frac{\bar{p}}{\bar{Z}}\right)\right]^{-1} \quad \text{(C-9)}$$

Substituting the values from Equations (C-4), (C-6) and (C-7) in Equation (C-9), it follows

$$\frac{\partial \bar{p}_p}{\partial t} = -\frac{2\, p_i\, q}{Z_i\, G \mu\, \bar{c}_g} \quad \text{(C-10)}$$

At this point, it is appropriate to introduce the definition of pseudotime for gas;

$$t_{ca} = \int \frac{dt}{\mu\, \bar{c}_g} \quad \text{(C-11)}$$

$$\frac{\partial t_{ca}}{\partial t} = \frac{1}{\mu\, \bar{c}_g} \quad \text{(C-12)}$$

Use the chain rule

$$\frac{\partial \bar{p}_p}{\partial t_{ca}} = \frac{\partial \bar{p}_p}{\partial t}\left(\frac{\partial t_{ca}}{\partial t}\right)^{-1} \quad \text{(C-13)}$$

$$\frac{\partial \bar{p}_p}{\partial t_{ca}} = -\frac{2\, p_i\, q}{G\, Z_i} \quad \text{(C-14)}$$

Assuming a constant rate q and integrating with appropriate limits

$$p_{p_i} - \bar{p}_p = \frac{2\, p_i\, q\, t_{ca}}{G\, Z_i} \quad \text{(C-15)}$$

Also recognizing that

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$$G = \frac{\phi A h p_i T_{st}}{Z_i p_{st} T} \quad \text{(C-16)}$$

Multiplying both sides of Equation (C-15) by *(kh/1.417qT)* and manipulating yields

$$\frac{kh}{1.417qT}(p_{p_i} - \overline{p}_p) = 2\pi \frac{2.637 \times 10^{-4} \times 24k\, t_{ca}}{\phi A} \quad \text{(C-17)}$$

Combining Equations C-1 and C-17 results in the Dynamic Material Balance Equation.

$$\overline{p}_p = p_{pwf} + q b_{pss} \quad \text{(C-18)}$$

where,

$$b_{pss} = \frac{1.417 \times 10^6 T}{kh}\left[\ln\left(\frac{r_e}{r_{wa}}\right) - \frac{3}{4}\right] \quad \text{(C-19)}$$

The above definition of b_{pss} applies to a vertical well in the center of a circular reservoir. Similar definitions, in terms of shape factors, can be developed for rectangular reservoirs.

The value of b_{pss} for a gas system is obtained from combining Equation C-1 with the definition of pseudotime.

$$p_{pi} - p_{pwf} = \frac{24 \times 2348 \times T \times q \times t_{ca}}{\pi \times \phi \times \mu_i \times c_{g_i} \times r_e^2 \times h} + \frac{1.417 \times 10^6 \times q \times T}{k \times h} \times \left[\ln\left(\frac{r_e}{r_{wa}}\right) - \frac{3}{4}\right] \quad \text{(C-20)}$$

This equation shows that a Cartesian plot of $(p_{pi}-p_{pwf}/q)$ versus t_{ca} will yield a straight line with an intercept of b_{pss}.

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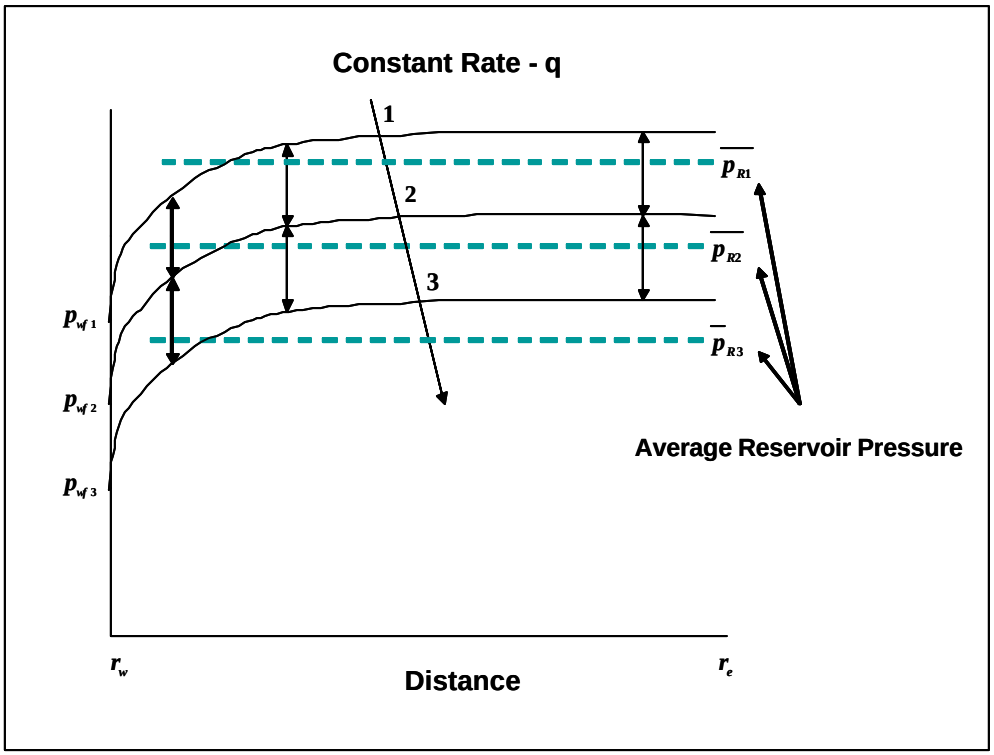

Figure 1: Pressure Drop in a Reservoir as a function of Radial Distance and Time During Boundary Dominated Flow

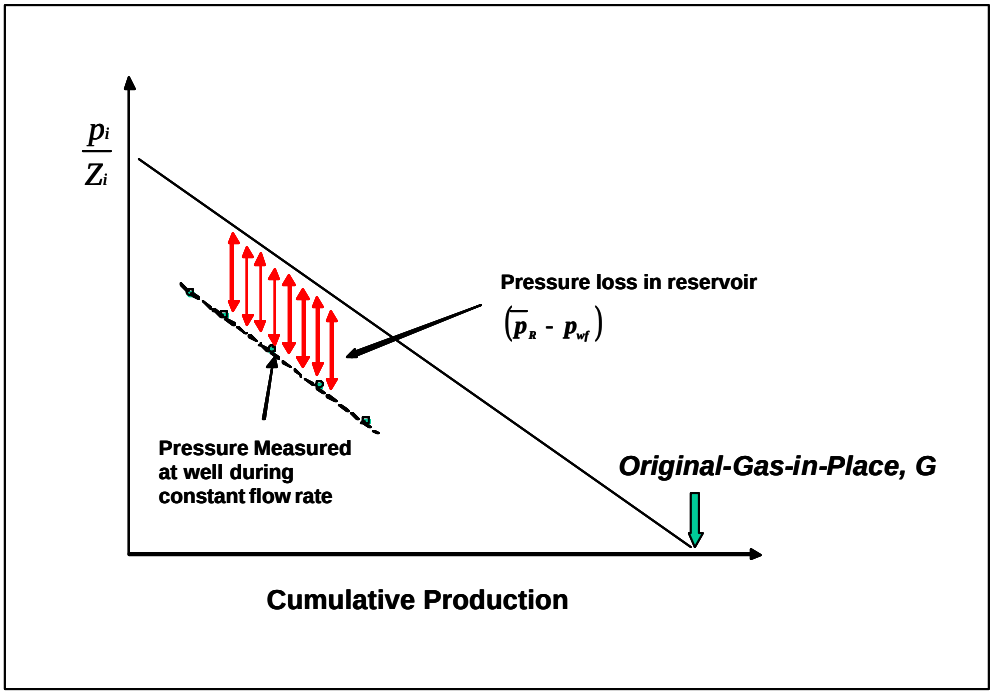

Figure 2: The Flowing P/Z Plot at Constant Rate Production

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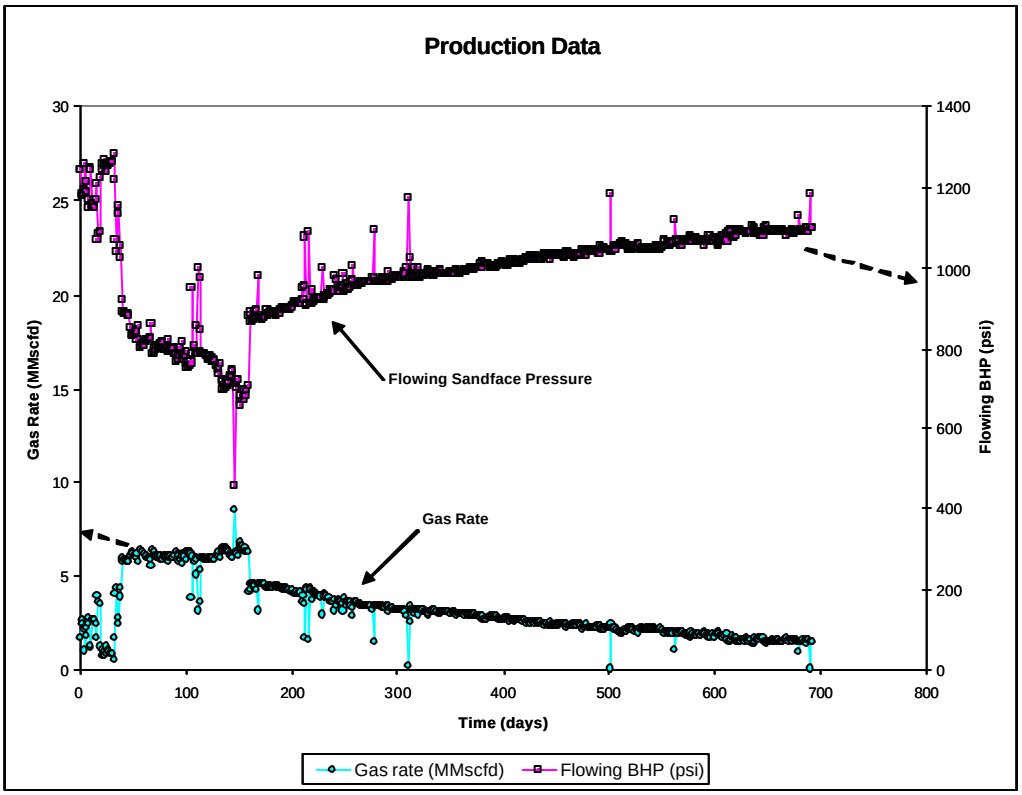

Figure 3: Production Data

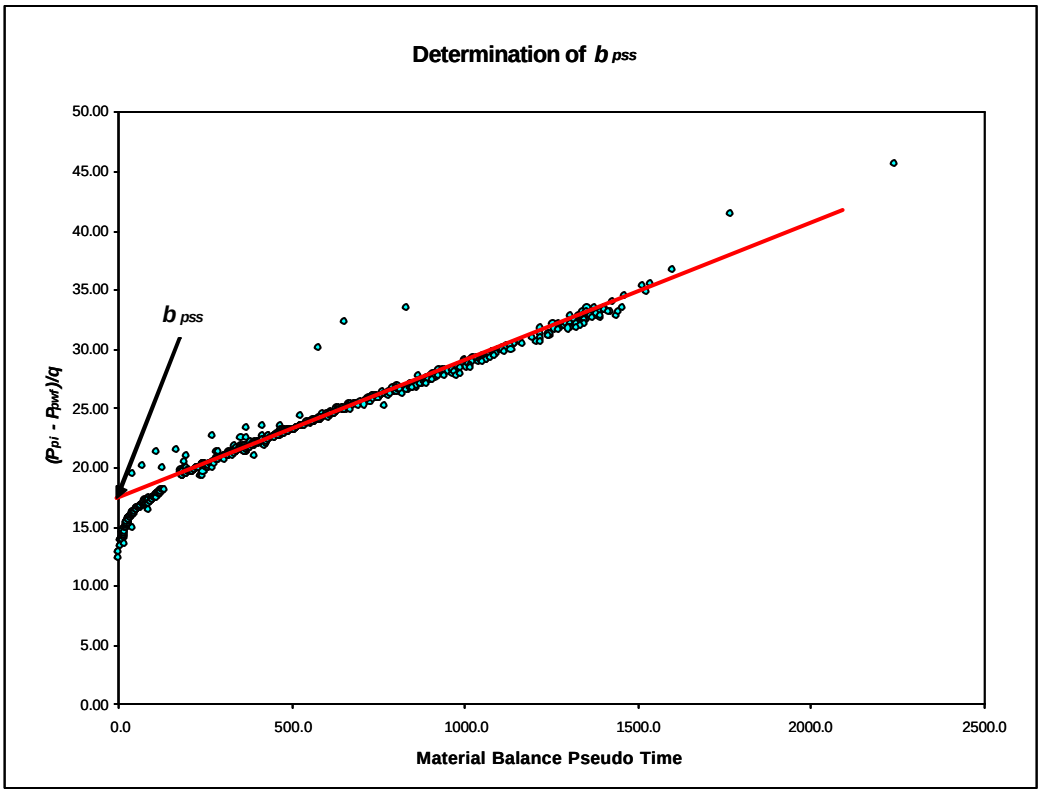

Figure 4: Determination of b_{pss}

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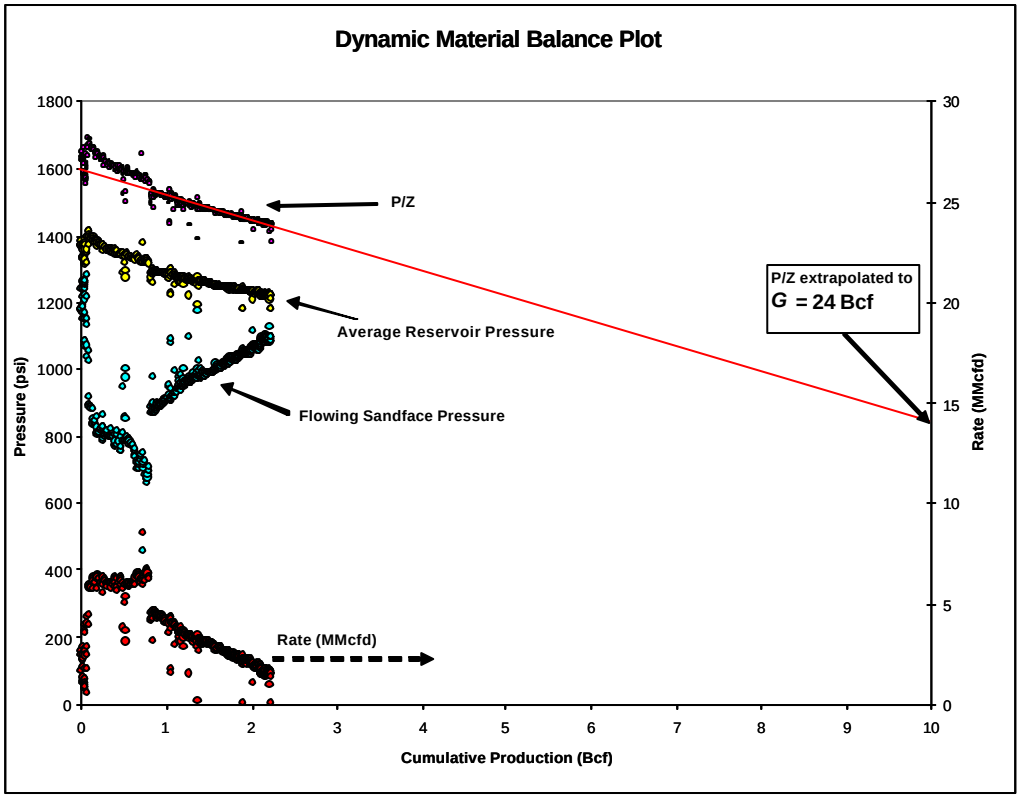

Figure 5: Dynamic Material Balance Plot

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